Test Series Code

07

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | DROPPER | **M. MARKS: 720**

DATE: 08/03/2026

ANSWER KEY

1.	(3)	46.	(1)	91.	(4)	136.	(3)
2.	(3)	47.	(2)	92.	(3)	137.	(3)
3.	(2)	48.	(1)	93.	(2)	138.	(3)
4.	(4)	49.	(1)	94.	(2)	139.	(3)
5.	(2)	50.	(1)	95.	(3)	140.	(4)
6.	(1)	51.	(2)	96.	(3)	141.	(2)
7.	(2)	52.	(3)	97.	(2)	142.	(3)
8.	(3)	53.	(2)	98.	(2)	143.	(3)
9.	(2)	54.	(3)	99.	(4)	144.	(2)
10.	(2)	55.	(4)	100.	(4)	145.	(1)
11.	(3)	56.	(4)	101.	(2)	146.	(2)
12.	(3)	57.	(3)	102.	(4)	147.	(4)
13.	(1)	58.	(3)	103.	(2)	148.	(1)
14.	(1)	59.	(3)	104.	(2)	149.	(4)
15.	(1)	60.	(2)	105.	(2)	150.	(2)
16.	(4)	61.	(2)	106.	(1)	151.	(2)
17.	(1)	62.	(3)	107.	(2)	152.	(3)
18.	(2)	63.	(2)	108.	(3)	153.	(4)
19.	(2)	64.	(4)	109.	(1)	154.	(3)
20.	(2)	65.	(2)	110.	(3)	155.	(3)
21.	(2)	66.	(3)	111.	(2)	156.	(2)
22.	(1)	67.	(2)	112.	(3)	157.	(1)
23.	(1)	68.	(4)	113.	(1)	158.	(2)
24.	(3)	69.	(2)	114.	(2)	159.	(1)
25.	(1)	70.	(1)	115.	(1)	160.	(4)
26.	(3)	71.	(3)	116.	(1)	161.	(2)
27.	(3)	72.	(1)	117.	(3)	162.	(4)
28.	(2)	73.	(3)	118.	(1)	163.	(1)
29.	(3)	74.	(3)	119.	(4)	164.	(2)
30.	(3)	75.	(2)	120.	(3)	165.	(3)
31.	(4)	76.	(2)	121.	(2)	166.	(4)
32.	(1)	77.	(4)	122.	(3)	167.	(1)
33.	(2)	78.	(3)	123.	(3)	168.	(3)
34.	(3)	79.	(2)	124.	(4)	169.	(1)
35.	(3)	80.	(3)	125.	(3)	170.	(2)
36.	(3)	81.	(3)	126.	(2)	171.	(1)
37.	(3)	82.	(2)	127.	(1)	172.	(4)
38.	(2)	83.	(3)	128.	(3)	173.	(3)
39.	(2)	84.	(3)	129.	(2)	174.	(3)
40.	(2)	85.	(1)	130.	(3)	175.	(2)
41.	(1)	86.	(3)	131.	(4)	176.	(3)
42.	(1)	87.	(3)	132.	(2)	177.	(3)
43.	(3)	88.	(1)	133.	(3)	178.	(2)
44.	(2)	89.	(1)	134.	(2)	179.	(1)
45.	(2)	90.	(2)	135.	(4)	180.	(2)

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Hints & Solutions

1. (3)

Since the resistance of the coil (R) is directly proportional to the length of the wire (and thus the number of turns), the resistance also increases by 100%.

$$V_s' = \frac{\text{New } I_s}{\text{New } R} = \frac{2 \times I_s}{2 \times R} = \frac{I_s}{R} = V_s$$

(NCERT Ed. 2025-26, 12th, Page No. 131)

2. (3)

$$\tau = k\theta \Rightarrow k = \frac{\tau}{\theta}$$

Dimension of Torque $(\tau) = [ML^2T^{-2}]$

Dimension of Angle $(\theta) =$ Dimensionless

$$\therefore [k] = [ML^2T^{-2}]$$

(NCERT Ed. 2025-26, 12th, Page No. 134)

3. (2)

$$m = NIA = NI(\pi r^2)$$

$$m = 100 \times \left(\frac{10}{\pi}\right) \times \pi(0.1)^2$$

$$m = 100 \times 10 \times 0.01 = 10 \text{ Am}^2 \text{ (Exact)}$$

$$\frac{1}{2} I_{inertia} \omega^2 = mB(\cos 0° - \cos 90°)$$

$$\frac{1}{2}(0.1)\omega^2 = 10 \times 2 \times (1 - 0)$$

$$0.05\omega^2 = 20$$

$$\omega^2 = \frac{20}{0.05} = 400 \Rightarrow \omega = 20 \text{ rad/s}$$

(NCERT Ed. 2025-26, 12th, Page No. 127)

4. (4)

Top Left Wire vs. Wire X: Both carry current into the page. They attract. The force on X is directed towards the top-left wire (↖).

Top Right Wire vs. Wire X: Top is out, X is into. They repel. The force on X is directed away from the top-right wire (down and to the left, ↙).

The vertical components of these two forces cancel out (one pulls up, one pushes down), while the horizontal components add up pointing to the left.

(NCERT Ed. 2025-26, 12th, Page No. 123)

5. (2)

Typical domain size is $1\,\text{mm}$ and the domain contains about 10^{11} atoms in a ferromagnetic material.

(NCERT Ed. 2025-26, 12th, Page No. 149)

6. (1)

Angle with normal $\theta = 90° - 60° = 30°$

$$E = NBA\omega\sin(\theta)$$

$$E = 50 \times 1 \times 0.2 \times 2 \times \sin(30°)$$

$$E = 20 \times 0.5$$

E = 10 V

(NCERT Ed. 2025-26, 12th, Page No. 171)

7. (2)

The S-pole moves towards the coil. To oppose this, the coil must repel the magnet by forming a South Pole at the end q.

A South Pole corresponds to a clockwise current (when viewed from the right).

Clockwise circulation means current leaves q, goes through the external wire, and enters p.

(NCERT Ed. 2025-26, 12th, Page No. 160)

8. (3)

Condition $T = 0 \Rightarrow \frac{mv^2}{l}$

$$= \text{mgsin}\theta \Rightarrow v = \sqrt{gl\sin\theta}$$

From energy conservation: $v_0 = \sqrt{gl(2 + 3\sin\theta)}$

$$\frac{v}{v_0} = \sqrt{\frac{\sin\theta}{2 + 3\sin\theta}}$$

$$\theta = 60^\circ: \text{ Ratio} = \sqrt{\frac{\frac{\sqrt{3}}{2}}{2 + \frac{3\sqrt{3}}{2}}} = \sqrt{\frac{\sqrt{3}}{4 + 3\sqrt{3}}}$$

(NCERT Ed. 2025-26, 11th, Page No. 79)

9. (2)

Frequency $f = \frac{qB}{2\pi m} = \frac{B}{2\pi}\left(\frac{e}{m}\right)$

$$f = \frac{3.57 \times 10^{-2} \times 1.76 \times 10^{11}}{2 \times 3.14}$$

$$f \approx \frac{6.28 \times 10^9}{6.28} = 10^9 \text{ Hz} = 1 \text{ GHz}$$

(NCERT Ed. 2025-26, 12th, Page No. 112)

[Contd...

10. (2)

Length $L = N \cdot 2\pi R \Rightarrow R = \dfrac{L}{2\pi N}$

Field $B = \dfrac{\mu_0 NI}{2R} = \dfrac{\mu_0 NI}{2(L/2\pi N)} = \dfrac{\mu_0 \pi I}{L} N^2$

So, $B \propto N^2$

$\dfrac{B'}{B} = \dfrac{n^2}{N^2} \Rightarrow B' = \dfrac{n^2}{N^2} B$

(NCERT Ed. 2025-26, 12th, Page No. 117)

11. (3)

$B_c = \dfrac{\mu_0 I}{2R}.$

$B_a = \dfrac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}.$

Substitute $x = \sqrt{3}R$: $R^2 + (\sqrt{3}R)^2 = 4R^2$.

Denominator becomes $2(4R^2)^{3/2} = 2(8R^3) = 16R^3$.

$B_a = \dfrac{\mu_0 I R^2}{16R^3} = \dfrac{\mu_0 I}{16R} = \dfrac{1}{8}\left(\dfrac{\mu_0 I}{2R}\right) = \dfrac{1}{8} B_c.$

Ratio $B_c : B_a = 8:1.$

(NCERT Ed. 2025-26, 12th, Page No. 116)

12. (3)

As the magnet falls, the changing magnetic flux induces eddy currents in the copper tube. These currents create an upward magnetic force that opposes gravity. When this retarding force equals the weight of the magnet ($F_m = mg$), the net force becomes zero, and the magnet achieves terminal velocity. Thus, the reason correctly explains the assertion.

(NCERT Ed. 2025-26, 12th, Page No. 160)

13. (1)

$T = 2\pi\sqrt{\dfrac{I}{MB}}$

Cut $\perp$ to length: Mass $\dfrac{m}{2}$, Length $\dfrac{L}{2}$

New Moment of Inertia

$I' = \dfrac{(m/2)(L/2)^2}{12} = \dfrac{1}{8}\dfrac{mL^2}{12} = \dfrac{I}{8}.$

New Magnetic Moment $M' = \dfrac{M}{2}.$

$T' = 2\pi\sqrt{\dfrac{I/8}{(M/2)B}} = 2\pi\sqrt{\dfrac{I}{4MB}} = \dfrac{1}{2}T.$

(NCERT Ed. 2025-26, 12th, Page No. 139)

14. (1)

Inside ($r < R$): Magnetic field $B \propto r$.

At $r = R/3, B_1 = \dfrac{\mu_0 I(R/3)}{2\pi R^2} = \dfrac{\mu_0 I}{6\pi R}$

Outside ($r > R$): Magnetic field $B \propto \dfrac{1}{r}$.

At $r = 3R, B_2 = \dfrac{\mu_0 I}{2\pi(3R)} = \dfrac{\mu_0 I}{6\pi R}.$

Ratio: $\dfrac{B_1}{B_2} = 1:1$

(NCERT Ed. 2025-26, 12th, Page No. 120)

15. (1)

From Faraday's Law in integral form:

$\oint \vec{E} \cdot d\vec{l} = \left|\dfrac{d\phi}{dt}\right|$

$E(2\pi r) = \dfrac{d}{dt}(B\pi r^2) = \pi r^2 \dfrac{dB}{dt}$

$E(2\pi r) = \pi r^2 \alpha$

$E = \dfrac{\alpha r}{2}$

(NCERT Ed. 2025-26, 12th, Page No. 163)

16. (4)

In option 4,

$3.6 \rightarrow 2$ s.f., $1.75 \rightarrow 3$ s.f.

so final result should have 2 s.f.

$\dfrac{3.6}{1.75} = 2.057 = 2.1$ m/s

(NCERT Ed. 2025-26, 11th, Page No. 5)

17. (1)

$W = MB(1 - \cos 60°) = MB(1 - 0.5)$

$= \dfrac{MB}{2} \Rightarrow MB = 2W.$

Torque $\tau = MB\sin 60^\circ = MB\dfrac{\sqrt{3}}{2}.$

Substitute MB: $\tau = (2W)\dfrac{\sqrt{3}}{2} = \sqrt{3}W$.

(NCERT Ed. 2025-26, 12th, Page No. 139)

18. (2)

For levitation, magnetic force per unit length = weight per unit length.

$I_{wire} B = \lambda g$

$I_{wire}(\mu_0 n I_{sol}) = \lambda g$

$10 \times (4\pi \times 10^{-7} \times 500 \times I_{sol}) = (2\pi \times 10^{-3}) \times 10$

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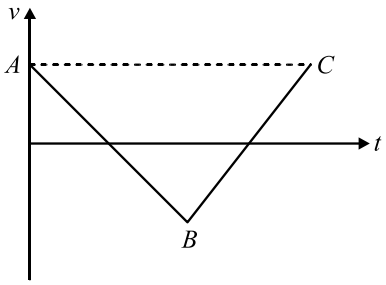
$10\times(2\pi\times10^{-4})\times I_{sol}=2\pi\times10^{-2}$

$2\pi\times10^{-3}\times I_{sol}=2\pi\times10^{-2}$

$I_{sol}=10$ A

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19. (2)



From A to B $\rightarrow$ slope is negative and constant,

$\therefore$ acceleration is negative and constant.

From B to C $\rightarrow$ Slope is positive and constant,

$\therefore$ acceleration is positive and constant.

(NCERT Ed. 2025-26, 11th, Page No. 16)

20. (2)

Diamagnetic substances are repelled by strong magnetic fields. Therefore, the liquid is pushed away from the field region, causing the level in that arm to be depressed (fall).

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21. (2)

Relative permeability $\mu_r=\dfrac{B_{medium}}{B_{vacuum}}=\dfrac{500B_0}{B_0}=500$

Since $\mu_r=1+\chi$, we get $\chi=\mu_r-1=500-1=499$

(NCERT Ed. 2025-26, 12th, Page No. 146)

22. (1)

Both liquids have equal masses therefore liquid with density ρ is filled to a height $\dfrac{2H}{3}$ and liquid with density 2ρ is filled to a height $\dfrac{H}{3}$.

Force at the base = Pressure at the base × Area.

$=\left[\rho g\left(\dfrac{2H}{3}\right)+2\rho g\left(\dfrac{H}{3}\right)\right]\times A$

$=\dfrac{4\rho gHA}{3}$

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23. (1)

For a short bar magnet, the magnetic field at the equatorial position is given by:

$B=\dfrac{\mu_0}{4\pi}\cdot\dfrac{M}{r^3}$

$B=10^{-7}\times\dfrac{0.4}{(0.2)^3}$

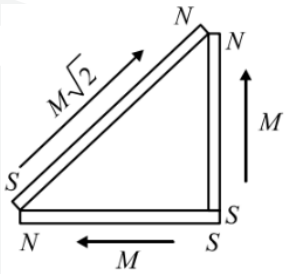
$B=10^{-7}\times\dfrac{0.4}{0.008}$

$B=10^{-7}\times50$

$B=5\times10^{-6}$ T

(NCERT Ed. 2025-26, 12th, Page No. 140)

24. (3)



Resultant of these three dipole moments will be

$M_{net}=\sqrt{(M\sqrt{2})^2+(M\sqrt{2})^2}=2M$

(NCERT Ed. 2025-26, 12th, Page No. 141)

25. (1)

$\tau=\dfrac{L}{R}$

$V_L=Ve^{-t/\tau}$

$V_R=V(1-e^{-t/\tau})$

$\tau=\dfrac{4}{10}=0.4$ s

$Ve^{-t/\tau}=V(1-e^{-t/\tau})$

$e^{-t/\tau}=1-e^{-t/\tau}$

$2e^{-t/\tau}=1\Rightarrow e^{t/\tau}=2$

$\dfrac{t}{\tau}=\ln 2$

$t=0.4\times0.69=0.28$ s

(NCERT Ed. 2025-26, 12th, Page No. 168)

26. (3)

$M=k\sqrt{L_1L_2}$

$k_{max}=1$

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[Contd...

27. (3)

Only statement B is correct. When magnetic flux changes at greater rate, induced emf is large. The induced charge does not depend on the total change of magnetic flux only.

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28. (2)

The total charge flow ΔQ depends only on the change in flux and resistance, not time.

$$\Delta Q = \frac{n \cdot \Delta\phi}{R}$$

$$\Delta Q = \frac{50 \times (2 \times 10^{-2})}{10}$$

$$\Delta Q = \frac{1}{10} = 0.1 \text{ C}$$

(NCERT Ed. 2025-26, 12th, Page No. 158)

29. (3)

$$V_A - iR + \varepsilon - L\frac{di}{dt} = V_B$$

$$V_A - (2 \times 3) + 7 - (12 \times 10^{-3})\frac{di}{dt} = V_B$$

$$V_A - V_B = -1 + (12 \times 10^{-3})\frac{di}{dt}$$

$$1 = 12 \times 10^{-3}\frac{di}{dt}$$

$$\frac{di}{dt} = \frac{1000}{12} = \frac{250}{3} \text{ A/s}$$

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30. (3)

Emf induced $= Bv\ell$

$= 0.4 \times 10 \times 1 = 4$ volt

Current induced in rod $= \frac{e}{R} = \frac{4}{2} = 2\text{A}$.

$F = Bi\ell \sin 90 = 0.4 \times 2 \times 1 = 0.8$ N

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31. (4)

$\Delta f = 526 - 512 = 14$ Hz

As $\Delta f > 10$ Hz, so human ears cannot hear the beats.

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32. (1)

Parallel Velocity:

$v_{\parallel} = v\cos 60^{\circ} = 2 \times 10^4 \times 0.5 = 10^4$ m/s

Time Period:

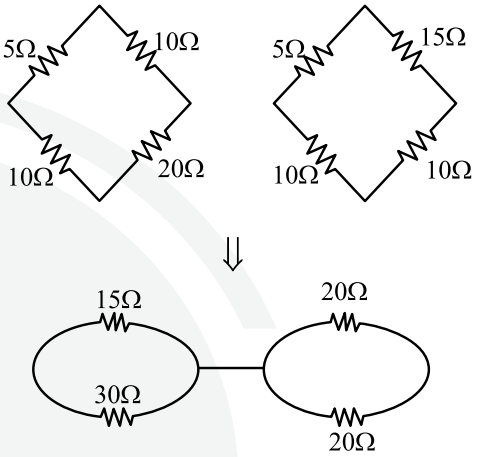
$$T = \frac{2\pi}{(q/m)B} = \frac{2\pi}{10^8 \times 2\pi \times 10^{-2}} = 10^{-6} \text{ s}$$

Pitch: $P = v_{\parallel} \times T = 10^4 \times 10^{-6} = 10^{-2}$ m $= 1$ cm

(NCERT Ed. 2025-26, 12th, Page No. 113)

33. (2)

As left combination is balanced wheatstone bridge.



$R_{eq} = 10 + 10 = 20\Omega$

$$i = \frac{20}{20} = 1 \text{ A}$$

(NCERT Ed. 2025-26, 12th, Page No. 100)

34. (3)

For adiabatic process $B = \gamma P$

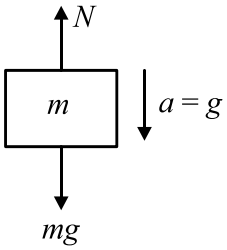
Hence, Bulk modulus depend on 'γ' of gas.

For isobaric process, $\Delta P = 0$

$$B = \frac{-V\Delta P}{\Delta V} = 0$$

(NCERT Ed. 2025-26, 11th, Page No. 287)

35. (3)



$mg - N = ma$

$N = mg - mg = 0$

As $N = 0$, so frictional force on the block is zero.

(NCERT Ed. 2025-26, 11th, Page No. 60)

[Contd...

36. (3)

EMF = $\frac{1}{2} \times B \times \omega \times l^2$

where l is the length from the center to the end $\left(l = \frac{L}{2}\right)$.

EMF = $\frac{1}{2} \times (1\text{ T}) \times (10\text{ rad/s}) \times (0.15\text{ m})^2$

EMF = $5 \times 0.0225 = 0.1125$ V

$|V_R - V_P| = 0.1125$ V

According to the right-hand rule on right side of rod, the force on positive charges in the rod is directed radially outward, to the center *R*. This means positive charges moves away from center *R*. Therefore, the center *R* is at the lowest potential.

$\therefore V_P = V_Q$ and V_R is at lowest potential.

(NCERT Ed. 2025-26, 12th, Page No. 164)

37. (3)

Gauss's law can be used to calculate electric field due to symmetrical charge distribution on charged conducting shell.

(NCERT Ed. 2025-26, 12th, Page No. 35)

38. (2)

1 → Isothermal expansion

2 → Adiabatic expansion

3 → Isothermal compression

4 → Adiabatic compression

(NCERT Ed. 2025-26, 11th, Page No. 238)

39. (2)

$L = mvr$

$L \propto \sqrt{r}$

$\frac{L_A}{L_B} = \sqrt{\frac{2R}{4R}} = \frac{1}{\sqrt{2}}$

(NCERT Ed. 2025-26, 11th, Page No. 137)

40. (2)

$F = 10^{-7} \times \frac{6 \times 5 \times 5}{(10^{-1})^4}$

$F = 10^{-7} \times \frac{150}{10^{-4}}$

$F = 150 \times 10^{-3}$ N

$F = 0.15$ N

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41. (1)

H = 1600 A/m

$\Phi = 2.4 \times 10^{-5}$ Wb

Area $A = 0.2 \times 10^{-4}$ m^2

Magnetic Induction $B = \frac{\Phi}{A} = \frac{2.4 \times 10^{-5}}{0.2 \times 10^{-4}} = 1.2$ T

Permeability $\mu = \frac{B}{H} = \frac{1.2}{1600} = 7.5 \times 10^{-4}$

Relative permeability

$\mu_r = \frac{\mu}{\mu_0} = \frac{7.5 \times 10^{-4}}{4\pi \times 10^{-7}} \approx 596.8$

Susceptibility $\chi = \mu_r - 1 \approx 596$

(NCERT Ed. 2025-26, 12th, Page No. 146)

42. (1)

Energy lost as heat $= \frac{1}{2}$ × work done by battery

$= \frac{1}{2} CV^2 = \frac{1}{2} \times 4 \times (10)^2$

= 200 μJ

(NCERT Ed. 2025-26, 12th, Page No. 74)

43. (3)

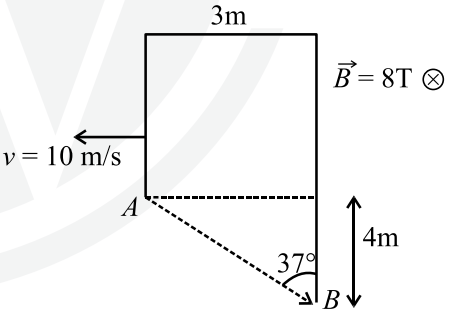
The net magnetic force on a current carrying loop in uniform magnetic field is zero.

$\vec{F}_{EFG} + \vec{F}_{GHE} = 0$

$\vec{F}_{GHE} = -\vec{T}$

(NCERT Ed. 2025-26, 12th, Page No. 133)

44. (2)



$\left|\vec{\ell}\right| = \sqrt{(3)^2 + (5)^2} = 5$m

emf $= \vec{\ell} \cdot (\vec{v} \times \vec{B})$

$= 5 \times 10 \times 8 \times \cos 37°$

$= 50 \times 8 \times \frac{4}{5} = 320$ volt

(NCERT Ed. 2025-26, 12th, Page No. 162)

45. (2)

$d\vec{B} = \frac{\mu_0 i}{4\pi} \left(\frac{d\vec{l} \times \vec{r}}{r^3} \right)$

(NCERT Ed. 2025-26, 12th, Page No. 113)

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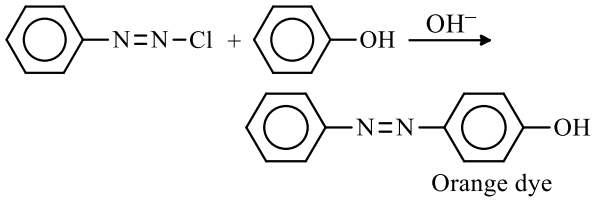
46. (1)

$$H-\underset{\underset{O}{\|}}{C}-H \xrightarrow[(ii)\ H_2O]{(i)\ \overset{\delta-}{CH_3}\overset{\delta+}{MgCl}} CH_3CH_2OH$$

(NCERT Ed. 2025-26 12th Part-II Page No. 201)

47. (2)

Coupling reaction:



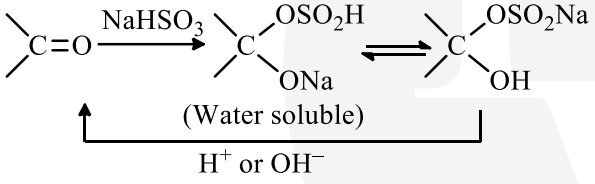
(NCERT Ed. 2025-26 12th Part-II Page No. 276)

48. (1)

In the given reaction dehydrohalogenation take place

(NCERT Ed. 2025-26 12th Part-II Page No. 179)

49. (1)



(NCERT Ed. 2025-26 12th Part-II Page No. 237)

50. (1)

- Isocyanide test (carbylamine reaction) is given by 1° amines (aliphatic or aromatic).

$$R-NH_2 + CHCl_3 + 3KOH \xrightarrow{\Delta} RNC + 3KCl + 3H_2O$$

- Aniline does not undergo Friedel-crafts alkyation reaction due to salt formation with $AlCl_3$, the Lewis acid, which is used as a catalyst.

(NCERT Ed. 2025-26 12th Part-II Page No. 271)

51. (2)

$$\Delta_f H^o = \Delta_{sub} H^o + \frac{1}{2}\Delta_{BE} H^o + \Delta_i H^o + \Delta_{eg} H^o + \Delta_{lattice} H^o$$

$$-617 = 161 + \frac{154}{2} + 520 + \Delta_{eg} H^o - 1047$$

$$\Delta_{eg} H^o = -328\ kJ\ mol^{-1}$$

(NCERT Ed. 2025-26 12th Part-I Page No. 155)

52. (3)

Reactivity due to steric and electronic effects:

$$\underset{(-M)}{3} > 2 > 4 > 1$$

(NCERT Ed. 2025-26 12th Part-II Page No. 236)

53. (2)

Basic nature:

$$\underset{\text{(Localised lone pair)}}{2} > \underset{\text{(H− effect)}}{3} > 1 > \underset{\text{(−M effect)}}{4}$$

(NCERT Ed. 2025-26 12th Part-II Page No. 269)

54. (3)

- $\overset{+7}{KMnO_4} \xrightarrow[3e^-]{OH^-} Mn^{4+}$
- $I^- \rightarrow IO_3^- + 6e^-$

$\therefore$ equivalent of $KMnO_4$ = equivalent of KI

$1 \times 3 = \text{mole} \times 6$

$n_{KI} = 0.5$

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55. (4)

- Clemmensen reduction:

$$C=O \xrightarrow[HCl]{Zn-Hg} CH_2$$

- Etard reaction:

$$CH_3 \xrightarrow[(ii)\ H_3O^+]{(i)\ CrO_2Cl_2,\ CS_2} CHO$$

- Wolff-Kishner reduction:

$$C=O \xrightarrow[(ii)\ KOH + \text{ethylene glycol}]{(i)\ N_2H_4} CH_2$$

- Stephen reaction:

$$R-CN \xrightarrow[(ii)\ H_3O^+]{(i)\ SnCl_2+HCl} R-CHO$$

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56. (4)

It is a mixed amine as it contains different alkyl groups.

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57. (3)

Total number of nodes = n – 1 = 4 – 1 = 3

(NCERT Ed. 2025-26 11th Part-I Page No. 59)

58. (3)

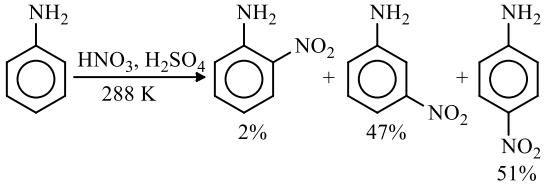
Aldol condensation reaction:

$$2CH_3CH_2CHO \xrightarrow[\Delta]{\text{dil. NaOH}} CH_3CH_2CH=\overset{CH_3}{\overset{|}{C}}-CHO$$

(NCERT Ed. 2025-26 12th Part-II Page No. 241)

[Contd...

59. (3)



(NCERT Ed. 2025-26 12th Part-II Page No. 273)

60. (2)

- $\underset{\text{Base}}{SO_4^{2-}} \xrightarrow{+H^+} \underset{\text{Conjugate acid}}{HSO_4^-}$
- Aqueous solution of NH_4Cl is acidic in nature due to cationic hydrolysis.

(NCERT Ed. 2025-26 11th Part-I Page No. 191)

61. (2)

Acidic strength:

2 > 3 > 4 > 1

(NCERT Ed. 2025-26 12th Part-II Page No. 251)

62. (3)

On the basis of extent of H-bonding, the order of boiling point for isomeric amines is: 1° > 2° > 3°.

(NCERT Ed. 2025-26 12th Part-II Page No. 265)

63. (2)

- CO_2: No. of atoms $= \frac{22}{44} \times 3 \times N_A = 1.5\, N_A$
- O_2: No. of atoms $= \frac{32}{32} \times 2 \times N_A = 2\, N_A$
- SO_2: No. of atoms $\frac{16}{64} \times 3 \times N_A = 0.75\, N_A$
- He: No. of atoms $= \frac{4}{4} \times 1 \times N_A = N_A$

(NCERT Ed. 2025-26 11th Part-I Page No. 18)

64. (4)

HVZ reaction is given by carboxylic acids having atleast one α-H atom.

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65. (2)

A and B are used for the identification of 1°, 2° and 3° amines.

(NCERT Ed. 2025-26 12th Part-II Page No. 271)

66. (3)

Alkane with odd number of carbon atoms are not obtain in good yield by wurtz reaction.

(NCERT Ed. 2025-26 11th Part-II Page No. 285)

67. (2)

$$Ph-MgCl \xrightarrow{CO_2} Ph-COOMgCl \xrightarrow{H_3O^+}$$

$$Ph-COOH \xrightarrow{NH_3,\,\Delta} Ph-CONH_2$$

(NCERT Ed. 2025-26 12th Part-II Page No. 252)

68. (4)

$$CH_3-\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-NH_2 \xrightarrow[(ii)\ H_2O]{(i)\ NaNO_2 + HCl} CH_3-\underset{CH_3}{\overset{CH_3}{\underset{|}{\overset{|}{C}}}}-OH$$

(NCERT Ed. 2025-26 12th Part-II Page No. 271)

69. (2)

Most of the worldwide production of phenol is from cumene.

(NCERT Ed. 2025-26 12th Part-II Page No. 202)

70. (1)

Decarboxylation:

$$C_6H_5COONa \xrightarrow[\Delta]{NaOH + CaO} C_6H_6$$

(NCERT Ed. 2025-26 12th Part-II Page No. 253)

71. (3)

Gabriel phthalimide reaction is used for the synthesis of 1° aliphatic amines.

(NCERT Ed. 2025-26 12th Part-II Page No. 264)

72. (1)

- $\begin{matrix} CH_3 \\ CH_3 \end{matrix}\!\!>\!CH-CH_2OH$: 2-Methylpropan-1-ol (isobutyl alcohol)
- $CH_3CH_2CH_2CH_2OH$: Butan-1-ol (n-butyl alcohol)

(NCERT Ed. 2025-26 11th Part-II Page No. 260)

73. (3)

Ketones do not give Tollen's test.

(NCERT Ed. 2025-26 12th Part-II Page No. 239)

74. (3)

The activating effect of $-NH_2$ group can be controlled by protecting the $-NH_2$ group by acetylation with acetic anhydride, then carrying out the desired substitution followed by hydrolysis of the substituted amide to substituted amine.

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[Contd...

75. (2)

Greater the number of methyl groups attached to a negatively charged carbon atom, lesser will be the stability of carbanion.

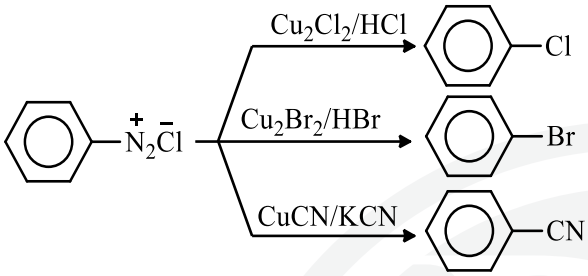
(NCERT Ed. 2025-26 11th Part-II Page No. 288)

76. (2)

Chemically Schiff's base is substituted imine.

(NCERT Ed. 2025-26 12th Part-II Page No. 238)

77. (4)



(NCERT Ed. 2025-26 12th Part-II Page No. 275)

78. (3)

CH_3Cl is most reactive due to least steric hinderance.

(NCERT Ed. 2025-26 12th Part-II Page No. 171)

79. (2)

Boiling point:

2 > 1 > 3 > 4

(NCERT Ed. 2025-26 12th Part-I Page No. 204)

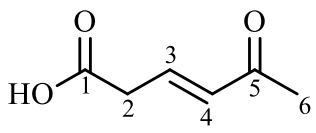
80. (3)

$$CH_3CH_2Br \xrightarrow{KCN} CH_3CH_2CN \xrightarrow{H_2O_2+NaOH}$$

$$CH_3CH_2CONH_2 \xrightarrow{LiAlH_4} CH_3CH_2CH_2NH_2$$

(NCERT Ed. 2025-26 12th Part-II Page No. 279)

81. (3)



5-Oxo hex-3-enoic acid

(NCERT Ed. 2025-26 11th Part-II Page No. 265)

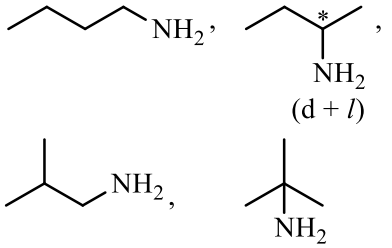
82. (2)

$$CH_3CH{=}CH_2 \xrightarrow[H_2O_2-NaOH]{B_2H_6} CH_3CH_2CH_2OH$$

$$\xrightarrow{Cu,\ \Delta} CH_3CH_2CHO$$

(NCERT Ed. 2025-26 12th Part-II Page No. 200)

83. (3)



(NCERT Ed. 2025-26 12th Part-II Page No. 261)

84. (3)

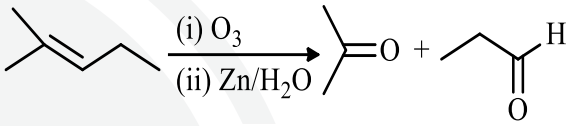
$$p_{total} = p_A^0 X_A + p_B^0 X_B$$

$$760 = 500(1 - X_B) + 800\,X_B$$

$$X_B = 0.87$$

(NCERT Ed. 2025-26 12th Part-I Page No. 10)

85. (1)



(NCERT Ed. 2025-26 12th Part-II Page No. 231)

86. (3)

Basic strength: 2° > 1° > 3°

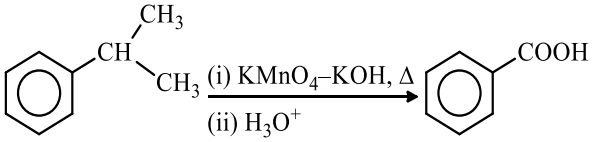
(NCERT Ed. 2025-26 12th Part-II Page No. 267)

87. (3)

Higher order (>3) reactions are rare it is due to low probability of simultaneous collision of all the reacting species.

(NCERT Ed. 2025-26 12th Part-I Page No. 68)

88. (1)



(NCERT Ed. 2025-26 12th Part-II Page No. 246)

89. (1)

RI is most reactive halide with amine.

(NCERT Ed. 2025-26 12th Part-II Page No. 270)

90. (2)

For weak electrolytes, conductivity decreases with dilution.

(NCERT Ed. 2025-26 12th Part-I Page No. 49)

[Contd...

91. (4)
Cytochrome *c* is a small protein attached to the outer surface of the inner mitochondrial membrane and acts as a mobile carrier for transfer of electrons between complex III and IV of the ETS.
(NCERT Ed. 2025-26, 11th, Page No. 160)

92. (3)
The organism shown is *Agaricus* (mushroom), which belongs to the class Basidiomycetes. In members of this class, sex organs are absent. Asexual spores are generally not found in Basidiomycetes; motile zoospores are characteristic of Phycomycetes. *Neurospora* (an Ascomycete) is the fungus used extensively in biochemical and genetic work. Plasmogamy is not immediately followed by karyogamy; instead, an intervening dikaryotic stage (n+n) occurs.
(NCERT Ed. 2025-26, 11th, Page No. 17)

93. (2)
When carbohydrates are used as substrate and are completely oxidised, the RQ will be 1, because equal amounts of CO_2 and O_2 are evolved and consumed, respectively, as shown in the equation below:

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + \text{Energy}$$

$$RQ = \frac{6CO_2}{6O_2} = 1.0$$

(NCERT Ed. 2025-26, 11th, Page No. 164)

94. (2)
Plants, unlike animals, have no specialised organs for gaseous exchange but they have stomata and lenticels for this purpose. There are several reasons why plants can get along without respiratory organs. First, each plant part takes care of its own gas-exchange needs. Second, plants do not present great demands for gas exchange. Roots, stems and leaves respire at rates far lower than animals do. Only during photosynthesis are large volumes of gases exchanged and each leaf is well adapted to take care of its own needs during these periods. When cells photosynthesise, availability of O_2 is not a problem in these cells since O_2 is released within the cell. Third, the distance that gases must diffuse even in large, bulky plants is not great. Each living cell in a plant is located quite close to the surface of the plant.
(NCERT Ed. 2025-26, 11th, Page No. 153)

95. (3)
In epigynous flowers, the margin of thalamus grows upward enclosing the ovary completely and getting fused with it, the other parts of flower arise above the ovary. Hence, the ovary is said to be inferior as in flowers of guava and cucumber, and the ray florets of sunflower.
China rose — Superior ovary
Plum — Half-inferior ovary
Mango — Superior ovary
(NCERT Ed. 2025-26, 11th, Page No. 63)

96. (3)
The F_1 headpiece is a peripheral membrane protein complex and contains the site for synthesis of ATP from ADP and inorganic phosphate. F_0 is an integral membrane protein complex that forms the channel through which protons cross the inner membrane. The passage of protons through the channel is coupled to the catalytic site of the F_1 component for the production of ATP. For each ATP produced, $4H^+$ pass through F_0 from the intermembrane space to the matrix down the electrochemical proton gradient.
(NCERT Ed. 2025-26, 11th, Page No. 161)

97. (2)
The quasi-fluid nature of lipid enables lateral movement of proteins within the overall bilayer. This ability to move within the membrane is measured as its fluidity.
(NCERT Ed. 2025-26, 11th, Page No. 94)

98. (2)
The complete combustion of glucose, which produces CO_2 and H_2O as end products, yields energy most of which is given out as heat.
(NCERT Ed. 2025-26, 11th, Page No. 155)

99. (4)
The production of cleistogamous flowers results in inbreeding depression because these flowers never open, ensuring that only self-pollination occurs.
(NCERT Ed. 2025-26, 12th, Page No. 12)

100. (4)
Complex II (Succinate dehydrogenase) does not pump any proton.
(NCERT Ed. 2025-26, 11th, Page No. 160)

101. (2)
Centrioles are made up of nine evenly spaced peripheral fibrils of tubulin protein. Each of the peripheral fibril is a triplet. The adjacent triplets are also linked. The central part of the proximal region of the centriole is also proteinaceous and called the hub, which is connected with tubules of the peripheral triplets by radial spokes made of protein. In centriole arrangement of microtubules is referred to as 9 (triplets) + 0 array.
(NCERT Ed. 2025-26, 11th, Page No. 100)

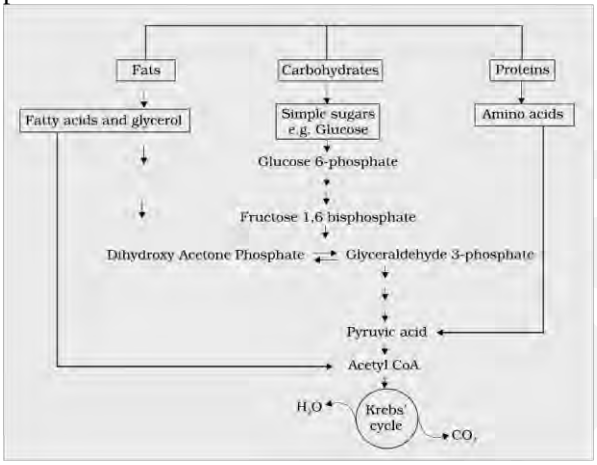
102. (4)

Glucose and fructose are phosphorylated to give rise to glucose-6- phosphate by the activity of the enzyme hexokinase. This phosphorylated form of glucose then isomerises to produce fructose-6-phosphate.

(NCERT Ed. 2025-26, 11th, Page No. 156)

103. (2)

Acetyl CoA is the common metabolite that serves as the entry point into the Krebs' cycle for the breakdown products of fats, carbohydrates, and proteins.



Interrelationship among metabolic pathways showing respiration mediated breakdown of different organic molecules to CO_2 and H_20

(NCERT Ed. 2025-26, 11th, Page No. 162)

104. (2)

The sporophyte of *Marchantia* (a liverwort) is differentiated into a foot, seta, and capsule. The sporophyte is developed from the zygote. It is the spore that germinates to produce the gametophyte. The sporophyte is not free-living; it is attached to and dependent on the photosynthetic gametophyte for nourishment. Sex organs (Antheridia and Archegonia) are produced on the gametophyte (specifically on antheridiophores and archegoniophores in *Marchantia*).

(NCERT Ed. 2025-26, 11th, Page No. 30)

105. (2)

During glycolysis, ATP is utilised at two steps: first in the conversion of glucose into glucose 6-phosphate and second in the conversion of fructose 6-phosphate to fructose 1, 6-bisphosphate.

(NCERT Ed. 2025-26, 11th, Page No. 156)

106. (1)

Lactate dehydrogenase	Pyruvate → Lactic acid
Invertase	Sucrose → Glucose + Fructose
Citrate synthase	Acetyl-CoA + OAA → Citric acid
Pyruvate dehydrogenase	Pyruvate → Acetyl-CoA

(NCERT Ed. 2025-26, 11th, Page No. 156-158)

107. (2)

All tissues on the innerside of the endodermis such as pericycle, vascular bundles and pith constitute the stele.

(NCERT Ed. 2025-26, 11th, Page No. 74)

108. (3)

Fatty acids would be broken down to acetyl CoA before entering the respiratory pathway when it is used as a substrate. But when the organism needs to synthesise fatty acids, acetyl CoA would be withdrawn from the respiratory pathway for it.

(NCERT Ed. 2025-26, 11th, Page No. 162)

109. (1)

When the organism needs to synthesise fatty acids, acetyl CoA would be withdrawn from the respiratory pathway for it. Hence, the respiratory pathway comes into the picture both during breakdown and synthesis of fatty acids. Similarly, during breakdown and synthesis of protein too, respiratory intermediates form the link. Breaking down processes within the living organism is catabolism, and synthesis is anabolism. Because the respiratory pathway is involved in both anabolism and catabolism, it would hence be better to consider the respiratory pathway as an amphibolic pathway rather than as a catabolic one.

(NCERT Ed. 2025-26, 11th, Page No. 162)

110. (3)

The RNA polymerase is only capable of catalysing the process of elongation. It associates transiently with initiation-factor (σ) and termination-factor (ρ) to initiate and terminate the transcription, respectively. Association with these factors alter the specificity of the RNA polymerase to either initiate or terminate.

(NCERT Ed. 2025-26, 12th, Page No. 94)

111. (2)

The role of oxygen is limited to the terminal stage of the process of aerobic respiration. Yet, the presence of oxygen is vital, since it drives the whole process by removing hydrogen from the system. Oxygen acts as the final hydrogen acceptor.

(NCERT Ed. 2025-26, 11th, Page No. 160)

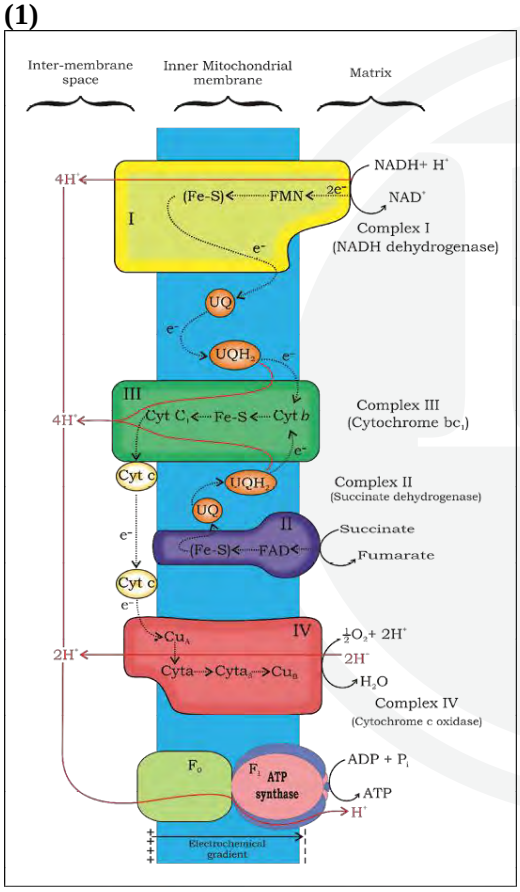
[Contd...

112. (3)

The second stage of prophase I called zygotene. During this stage, chromosomes start pairing together and this process of association is called synapsis. Such paired chromosomes are called homologous chromosomes. Electron micrographs of this stage, indicate that chromosome synapsis is accompanied by the formation of complex structure called synaptonemal complex. The complex formed by a pair of synapsed homologous chromosomes is called a bivalent or a tetrad. However, these are more clearly visible at the next stage. The first two stages of prophase I are relatively short-lived compared to the next stage that is pachytene.

(NCERT Ed. 2025-26, 11th, Page No. 126)

113. (1)



Electron Transport System (ETS)

(NCERT Ed. 2025-26, 11th, Page No. 160)

114. (2)

ATP acts as the energy currency of the cell. This energy trapped in ATP is utilised in various energy-requiring processes of the organisms, and the carbon skeleton produced during respiration is used as precursors for biosynthesis of other molecules in the cell.

(NCERT Ed. 2025-26, 11th, Page No. 154)

115. (1)

The correct sequence of events in the gymnosperm life cycle follows the progression from the formation of the ovule to the production of the seed.

The ovule (megasporangium) develops on the megasporophyll. One cell of the nucellus (megaspore mother cell) undergoes meiosis (reductional division) to form four haploid megaspores. One of the megaspores develops into a multicellular female gametophyte that bears two or more archegonia. After pollination, the pollen tube grows through the tissues of the nucellus toward the archegonia to deliver the male gametes. Following fertilization, the zygote develops into an embryo and the ovules develop into seeds (which are not covered).

(NCERT Ed. 2025-26, 11th, Page No. 33)

116. (1)

In both lactic acid and alcohol fermentation not much energy is released; less than seven per cent of the energy in glucose is released and not all of it is trapped as high energy bonds of ATP.

(NCERT Ed. 2025-26, 11th, Page No. 157)

117. (3)

For each ATP produced, $4H^+$ pass through F_0 from the mitochondrial intermembrane space to the matrix down the electrochemical proton gradient.

(NCERT Ed. 2025-26, 11th, Page No. 161)

118. (1)

In a typical angiosperm anther, the tapetum is the innermost wall layer that nourishes the developing pollen grains (microspores).

(NCERT Ed. 2025-26, 12th, Page No. 5)

119. (4)

While NADH + H^+ is produced during the initial steps of glycolysis, it is completely consumed during the subsequent fermentation process. Therefore, there is no net gain of NADH in alcoholic fermentation.

(NCERT Ed. 2025-26, 11th, Page No. 157)

120. (3)

Metabolic reactions can be demonstrated outside the body in cell-free systems. An isolated metabolic reaction(s) outside the body of an organism, performed in a test tube is neither living nor non-living. Hence, while metabolism is a defining feature of all living organisms without exception, isolated metabolic reactions *in vitro* are not living things but surely living reactions.

(NCERT Ed. 2022-23, 11th, Page No. 5)

121. (2)

The TCA cycle starts with the condensation of acetyl group with oxaloacetic acid (OAA) and water to yield citric acid. The reaction is catalysed by the enzyme citrate synthase and a molecule of CoA is released. Citrate is then isomerised to isocitrate. It is followed by two successive steps of decarboxylation, leading to the formation of α-ketoglutaric acid and then succinyl-CoA.

(NCERT Ed. 2025-26, 11th, Page No. 158)

[Contd...

122. (3)

The energy released by oxidation in respiration is not (or rather cannot be) used directly but is used to synthesise ATP, which is broken down whenever (and wherever) energy needs to be utilised. Hence, ATP acts as the energy currency of the cell. This energy trapped in ATP is utilised in various energy-requiring processes of the organisms, and the carbon skeleton produced during respiration is used as precursors for biosynthesis of other molecules in the cell.

(NCERT Ed. 2025-26, 11th, Page No. 154)

123. (3)

Floral formula for leguminosae:

$$\% \, ♂♀ \, K_{(5)} \, C_{1+2+(2)} \, A_{(9)+1} \, \underline{G}_1$$

(NCERT Ed. 2025-26, 11th, Page No. 79)

124. (4)

There can be a net gain of 38 ATP molecules during aerobic respiration of one molecule of glucose. NADH is oxidised to NAD^+ rather slowly in fermentation, however the reaction is very vigorous in case of aerobic respiration.

(NCERT Ed. 2025-26, 11th, Page No. 162)

125. (3)

Oxidation of one molecule of NADH gives rise to 3 molecules of ATP.

(NCERT Ed. 2025-26, 11th, Page No. 160)

126. (2)

The genetic code is nearly universal: for example, from bacteria to human UUU would code for Phenylalanine (phe). Some exceptions to this rule have been found in mitochondrial codons, and in some protozoans.

(NCERT Ed. 2025-26, 12th, Page No. 96)

127. (1)

Glycolysis is the only process in respiration for anaerobic organisms and is the first stage of respiration in all living organisms, including aerobic ones. The pathway occurs in the cytoplasm and does not require oxygen to break down glucose into pyruvic acid. Because it does not depend on oxygen, it can function in various organisms regardless of their oxygen requirements (aerobes or anaerobes).

(NCERT Ed. 2025-26, 11th, Page No. 155)

128. (3)

During the conversion of succinyl-CoA to succinic acid a molecule of GTP is synthesised. This is a substrate level phosphorylation. In a coupled reaction GTP is converted to GDP with the simultaneous synthesis of ATP from ADP.

(NCERT Ed. 2025-26, 11th, Page No. 159)

129. (2)

There is one step in glycolysis where $NADH + H^+$ is formed from NAD^+; this is when 3-phosphoglyceraldehyde (PGAL) is converted to 1, 3-bisphosphoglycerate (BPGA). Two redox-equivalents are removed (in the form of two hydrogen atoms) from PGAL and transferred to a molecule of NAD^+. PGAL is oxidised and with inorganic phosphate to get converted into BPGA.

(NCERT Ed. 2025-26, 11th, Page No. 156)

130. (3)

Complex IV refers to cytochrome *c* oxidase complex containing cytochromes *a* and a_3, and two copper centres.

(NCERT Ed. 2025-26, 11th, Page No. 160)

131. (4)

Lactic acid	3 carbons
Isocitrate	6 carbons
Fumarate	4 carbons
α-Ketoglutaric acid	5 carbons

(NCERT Ed. 2025-26, 11th, Page No. 157-159)

132. (2)

Glycolysis occurs in the cytoplasm of the cell and is present in all living organisms. The end product of glycolysis is pyruvic acid (two molecules). Acetyl CoA is formed later during oxidative decarboxylation inside the mitochondria.

(NCERT Ed. 2025-26, 11th, Page No. 155-156)

133. (3)

If fatty acids were to be respired, they would first be degraded to acetyl CoA and enter the pathway. Glycerol would enter the pathway after being converted to PGAL.

(NCERT Ed. 2025-26, 11th, Page No. 162)

134. (2)

When proteins are used as respiratory substrates, the Respiratory Quotient (RQ) is approximately 0.9, which means more oxygen is consumed than carbon dioxide is released.

(NCERT Ed. 2025-26, 11th, Page No. 164)

135. (4)

NADH dehydrogenase - complex I
Succinate dehydrogenase - complex II
Cytochrome bc_1 complex - complex III
Cytochrome *c* oxidase complex - Complex IV

(NCERT Ed. 2025-26, 11th, Page No. 160)

[Contd...

136. (3)

Adaptive ability is inherited and has a genetic basis. Existing life forms share similarities not only with modern organisms but also with those that existed millions of years ago. Biological and geological histories are closely correlated, and life actually appeared about 4 billion years ago, not 4.5 billion.

(NCERT Ed. 2025-26, 12th, Page No. 118)

137. (3)

Magnetic Resonance Imaging (MRI), not Computed Tomography (CT), that uses strong magnetic fields and non-ionising radiations to detect pathological changes. CT scans utilize X-rays to generate three-dimensional images of internal organs.

(NCERT Ed. 2025-26, 12th, Page No. 141)

138. (3)

Homo habilis	Brain size 650-800cc; first hominid
Homo erectus	Brain size 900cc
Neanderthal	Brain size 1400cc; buried dead
Australopithecines	Lived in East African grasslands; ate fruit

(NCERT Ed. 2025-26, 12th, Page No. 124)

139. (3)

Although diaphragms, cervical caps, and vaults are effective barriers, they do not prevent the transmission of STIs and AIDS. Only condoms are highly effective in providing protection against sexually transmitted infections.

(NCERT Ed. 2025-26, 12th, Page No. 44)

140. (4)

The universe is indeed about 13.8 billion years old, life appeared 500 million years after Earth's formation, and early Earth lacked an atmosphere. Statement B is wrong because Earth formed 4.5 billion years ago, and E is wrong because stellar distances are measured in light-years.

(NCERT Ed. 2025-26, 12th, Page No. 111)

141. (2)

Enzymes function by lowering the activation energy required for a reaction, which significantly increases the rate of that reaction. With the increase in substrate concentration, the velocity of the enzymatic reaction rises at first, and most enzymes are proteins that denature at high temperatures.

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142. (3)

Paleontological evidence for evolution is derived from the study of fossils, which are the remains of hard parts of life-forms found in sedimentary rock layers. These fossils represent extinct organisms and show how life-forms have varied over different geological time periods.

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143. (3)

Pheretima (Earthworm) is the correct choice as it is triploblastic, bilaterally symmetrical, and possesses a true coelom. *Taenia* is acoelomate, *Ascaris* is a pseudocoelomate, and *Adamsia* is a diploblastic organism with radial symmetry.

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144. (2)

Charles Darwin used the sail ship H.M.S. Beagle for his famous sea voyage around the world. Based on observations made during this journey, he concluded that existing life forms share similarities with each other and with extinct forms from the past.

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145. (1)

Evolution is considered a stochastic process because it is not deterministic; rather, it is based on random chance events in the environment and chance mutations occurring within organisms.

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146. (2)

While individual groups like Australian marsupials show adaptive radiation (divergent evolution), the comparison between marsupials and placental mammals is an example of convergent evolution. This occurs when different groups evolve similar features because they adapt to similar habitats.

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[Contd...

147. (4)

The ejaculatory duct is formed by the union of the ampulla of vas deferens and the duct of seminal vesicle. It passes through the prostate gland and opens into the prostatic urethra. Its function is to transport sperm and seminal fluid into the urethra during ejaculation.

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148. (1)

Mutation Theory	Hugo de Vries
Life originated from pre-existing life	Louis Pasteur
Use and Disuse of organs	Lamarck
Malay Archipelago	Alfred Wallace

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149. (4)

Industrial melanism is an example of all the listed factors. It represents evolution triggered by human (anthropogenic) industrial activity, demonstrates natural selection through camouflage, and is a classic case of directional selection where the population shifts toward one extreme phenotype.

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150. (2)

This phenomenon is best described as genetic drift. It refers to the change in allele frequency that occurs by chance in a small, isolated population; when this leads to the formation of a new species, it is specifically called the founder effect.

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151. (2)

The cerebellum is correctly paired with its function of coordinating complex body movements and maintaining balance and posture. Functions like temperature control belong to the hypothalamus, while the medulla contains vital centers for respiration and cardiovascular reflexes.

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152. (3)

The correct statement is that Neanderthals lived in near east and central Asia between 1,00,000–40,000 years ago, used hides for protection, and buried their dead. *Homo habilis* probably did not eat meat, and *Homo erectus* appeared about 1.5 million years ago.

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153. (4)

Oxytocin and Vasopressin (ADH) are the hormones synthesized by the hypothalamus and then transported to the neurohypophysis (posterior pituitary) for storage and release. The adenohypophysis (anterior pituitary) synthesizes hormones like TSH, LH, and ACTH.

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154. (3)

The diagram represents directional selection. This occurs when natural selection favors one extreme phenotype over the others, causing the peak of the distribution curve to shift in that specific direction rather than remaining at the mean.

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155. (3)

The ascending limb of the loop of Henle is the specific part of the nephron that is impermeable to water but allows the transport (active or passive) of electrolytes like NaCl. This unique property is essential for maintaining the high osmolarity of the medullary interstitium.

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156. (2)

Analysis of meteorite content revealed the presence of organic compounds similar to those found in Miller's experiment. This suggests that similar chemical evolution processes, involving the formation of organic molecules from inorganic constituents, are occurring elsewhere in space.

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157. (1)

Hugo de Vries proposed that mutations are random and directionless, leading to sudden speciation, whereas Darwin believed that evolution occurs through small, heritable variations that are directional and gradual.

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158. (2)

2000 mya	First cellular forms of life
500 mya	Invertebrates formed
350 mya	Jawless fish evolved
320 mya	Sea weeds and few plants

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[Contd...

159. (1)

Natural selection is based on facts like limited resources and population stability; Darwin was influenced by Malthus, and fitness is the result of adaptation. Alfred Russel Wallace was co-proposer of the theory of evolution by natural selection, not by use and disuse of organs. Theory of use and disuse of organs is proposed by Jean-Baptiste Lamarck. Example: Giraffe's neck became long due to repeated stretching. Also called Lamarckism.

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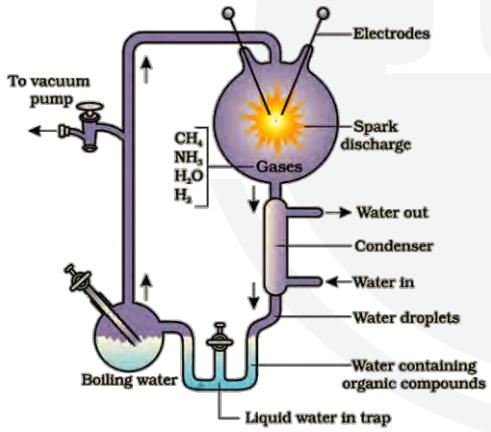
160. (4)

Saddle joint is found between the carpals and metacarpals of the human thumb. Other joints include the pivot joint between the atlas and axis, and the hinge joint in places like the elbow or knee.

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161. (2)

In Miller's experiment, the specific conditions created in the closed flask (X) were an electric discharge in the presence of CH_4, H_2, NH_3 and water vapor at 800 °C**.** This experiment successfully demonstrated the formation of amino acids from inorganic gases.



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162. (4)

Higher temperature is not favorable for oxyhaemoglobin formation; it actually promotes the dissociation of oxygen from hemoglobin. Formation at the alveolar site is favored by high pO_2, low pCO_2, and a lesser H^+ concentration.

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163. (1)

Sweet potato and Potato	Analogy (Root/Stem)
Lichens	Do not grow in polluted areas
Herbicide resistance	Anthropogenic evolution
Vertebrate Hearts	Homology

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164. (2)

The theory of special creation claims that diversity was always the same since creation and will remain the same in the future. It also asserts that all species were created exactly as they appear today and that the Earth is only 4000 years old.

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165. (3)

Embryological support was proposed by Heckel and disapproved by von Baer, homology indicates common ancestry, and biochemical similarities also point to a shared past. Fossils are hard parts, and gill slits aren't in all adult vertebrates.

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166. (4)

Bobcat	Tasmanian tiger cat
Lemur	Spotted cuscus
Anteater	Numbat
Flying squirrel	Flying phalanger

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167. (1)

Heroin (smack) is chemically identified as diacetylmorphine. It is produced through the acetylation of morphine, which is a natural substance extracted from the latex of the poppy plant, *Papaver somniferum*.

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168. (3)

Genital herpes, Hepatitis-B, and HIV infection are the three sexually transmitted infections that are currently not completely curable. Other STIs like syphilis and gonorrhea can be cured if they are detected early and treated with proper medical care.

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169. (1)

Coelacanth was caught in 1938, *T. rex* was 20 feet tall, dinosaurs died out 65 mya, and small reptiles from that era still exist. Statement A is incorrect because plants, not animals, were the first organisms to invade land.

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170. (2)

Oparin and Haldane proposed chemical evolution, suggesting that the first forms of life arose from pre-existing non-living organic molecules like RNA and proteins. They argued that life's appearance was preceded by the formation of diverse organic molecules from inorganic constituents.

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[Contd...

171. (1)

The first non-cellular forms of life originated about 3 billion years ago and could have been giant molecules like RNA or proteins. Statement II is incorrect because it was the first cellular forms of life that originated about 2000 million years ago.

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172. (4)

The gene pool remains constant in equilibrium, the sum of allelic frequencies is 1, and natural selection leads to stabilization when it favors the mean character. Statement C is wrong because migration must happen *multiple times* for gene flow, and D is incorrect as large population size and random mating are conditions required to maintain Hardy–Weinberg equilibrium, not factors that disturb it.

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173. (3)

The 'Panspermia' theory, which remains a favorite for some astronomers, suggests that units of life called spores were transferred from outer space to different planets, including Earth. This theory attempts to explain the origin of life as an extraterrestrial event.

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174. (3)

In a standard ECG, the T-wave represents the repolarisation of the ventricles, which is the process where the ventricles return from an excited state to their normal, resting state. The end of the T-wave specifically marks the end of the ventricular systole.

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175. (2)

Mesozoic Era	Birds and Reptiles
Proterozoic Era	Lower invertebrates
Cenozoic Era	Mammals
Paleozoic Era	Fish and Amphibia

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176. (3)

The agriculture began approximately 10,000 years ago, leading to the start of human settlements. *Ramapithecus* was more man-like, *Dryopithecus* was more ape-like, and pre-historic cave art developed about 18,000 years ago.

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177. (3)

During aestivation and hibernation, when the frog is in a state of dormancy, respiration occurs exclusively through the skin (cutaneous respiration). While they use lungs and the buccal cavity on land, these methods are not used during these dormant periods.

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178. (2)

Branching Descent, which, along with natural selection, is one of the two key concepts of the Darwinian Theory of Evolution. Darwin's finches represent adaptive radiation (not convergent), and saltation refers to a single-step large mutation.

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179. (1)

Areolar tissue is a type of loose connective tissue found beneath the skin that serves as a support framework for epithelium. It contains various cells, including fibroblasts (which produce fibers), macrophages, and mast cells, all embedded in a semi-fluid ground substance.

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180. (2)

It is a biological fact that the baby chimpanzee skull resembles the adult human skull more than its own adult form, but this physical similarity is not a direct result of mammals being viviparous.

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